ADMISSIBLE SUBCATEGORIES AND SEMI-ORTHOGONAL DECOMPOSITIONS VIA QUOTIENT EQUIVALENCES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 12192-12220, following the statement at 12169-12190.

Editorial note. The (1)-(2)-(3) equivalence is delegated to lemma-right-adjoint and lemma-left-adjoint, both fully proved in this chapter; the author's 'small detail omitted' is the convention matching of the perpendicular notation, not a load-bearing construction.

Editorial note. The dual half (B to D/A) is explicitly marked as dual to the executed argument, following the established dual-mirror counting practice.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 12195 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Right/left adjoint characterization of admissibility. lemma-right-adjoint (12048-12067) and lemma-left-adjoint (12117-12136), both same chapter, cited at 12193-12195. Their statements are exactly the equivalence of condition (1) (respectively (2)) with the triangle decomposition and orthogonality of condition (3).
- Adjoint fully faithful left quasi-inverse. Categories, Lemma categories-lemma-adjoint-fully-faithful (cross-chapter, cited around line 12204). v composed with the inclusion is the identity on A, making v-bar a left quasi-inverse of i-bar.
- Universal property of the quotient. quotient construction of D/B (same chapter section on the quotient). Factors v through D/B since its kernel is B.

Source reference. Original reference 07RB: focused-seeds/geometry/S9-der-summarize-admissible/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CQT: sources/stacks-derived/pinned-original-source.tex, source lines 12118–12118. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0CQS: sources/stacks-derived/pinned-original-source.tex, source lines 12049–12049. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let $\mathcal{D}$ be a triangulated category. Let $\mathcal{A} \subset \mathcal{D}$ and $\mathcal{B} \subset \mathcal{D}$ be subcategories. The following are equivalent*

(1) *$\mathcal{A}$ is right admissible and $\mathcal{B} = \mathcal{A}^\perp$,*
(2) *$\mathcal{B}$ is left admissible and $\mathcal{A} = {}^\perp\mathcal{B}$,*
(3) *$\mathrm{Hom}(A, B) = 0$ for all $A \in \mathcal{A}$ and $B \in \mathcal{B}$ and for every X in $\mathcal{D}$ there exists a distinguished triangle $A \to X \to B \to A[1]$ in $\mathcal{D}$ with $A \in \mathcal{A}$ and $B \in \mathcal{B}$.*

If this is true, then $\mathcal{A} \to \mathcal{D}/\mathcal{B}$ and $\mathcal{B} \to \mathcal{D}/\mathcal{A}$ are equivalences of triangulated categories, the right adjoint to the inclusion functor $\mathcal{A} \to \mathcal{D}$ is $\mathcal{D} \to \mathcal{D}/\mathcal{B} \to \mathcal{A}$, and the left adjoint to the inclusion functor $\mathcal{B} \to \mathcal{D}$ is $\mathcal{D} \to \mathcal{D}/\mathcal{A} \to \mathcal{B}$.

Proof. The equivalence between (1), (2), and (3) follows in a straightforward manner from Lemmas 0CQS and 0CQT (small detail omitted). Denote $v : \mathcal{D} \to \mathcal{A}$ the right adjoint of the inclusion functor $i : \mathcal{A} \to \mathcal{D}$. It is immediate that $\mathrm{Ker}(v) = \mathcal{A}^\perp = \mathcal{B}$. Thus v factors over a functor $\overline{v} : \mathcal{D}/\mathcal{B} \to \mathcal{A}$ by the universal property of the quotient. Since $v \circ i = \mathrm{id}_\mathcal{A}$ by Categories, Lemma 07RB we see that $\overline{v}$ is a left quasi-inverse to $\overline{i} : \mathcal{A} \to \mathcal{D}/\mathcal{B}$. We claim also the composition $\overline{i} \circ \overline{v}$ is isomorphic to $\mathrm{id}_{\mathcal{D}/\mathcal{B}}$. Namely, suppose we have X fitting into a distinguished triangle $A \to X \to B \to A[1]$ as in (3). Then $v(X) = A$ as was seen in the proof of Lemma 0CQS. Viewing X as an object of $\mathcal{D}/\mathcal{B}$ we have $\overline{i}(\overline{v}(X)) = A$ and there is a functorial isomorphism $\overline{i}(\overline{v}(X)) = A \to X$ in $\mathcal{D}/\mathcal{B}$. Thus we find that indeed $\overline{v} : \mathcal{D}/\mathcal{B} \to \mathcal{A}$ is an equivalence. To show that $\mathcal{B} \to \mathcal{D}/\mathcal{A}$ is an equivalence and the left adjoint to the inclusion functor $\mathcal{B} \to \mathcal{D}$ is $\mathcal{D} \to \mathcal{D}/\mathcal{A} \to \mathcal{B}$ is dual to what we just said. □